# FSK 176

# Examination

# Department of Physics, University of Pretoria

**Total time for the examination (Part 1 and Part 2): 3 hours**

**Examiner: Prof. JM Nel** **Moderator: Prof. WE Meyer**

**Please read the following carefully.**

1. Answer all the questions on lined or blank paper.
2. Once you are done, scan your answers into a pdf document and submit your answers on Gradescope.
3. This is an open book test - you are allowed to use your textbook when answering this test. You may not search the internet for solutions to the problems.
4. Where applicable you need to draw the necessary labelled diagrams to assist with answering the question - such as free body diagrams.
5. Your answer needs to be written in a systematic and logical fashion. Tell us what you are doing. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.**
6. **Units must be used when numbers are substituted else marks will be subtracted.**
7. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner.

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**Declaration:**

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment
In proceeding with this test, you are accepting the contents of the above declaration.

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**Constants and formulas**
$G = 6.674 \times 10^{-11}$ N·m$^2$/kg$^2$; $e = 1.602 \times 10^{-19}$ C; $k = 1.381 \times 10^{-23}$ J/K; $m_e = 9.109 \times 10^{-31}$ kg; $m_p = 1.673 \times 10^{-27}$ kg; $R = 8.314$ J/(mol·K); $N_A = 6.02 \times 10^{23}$ /mol; 1 atm = 101.3 kPa; 1 u = $1.6605677 \times 10^{-27}$ kg; $c = 3 \times 10^8$ m/s; $V_{\text{sphere}} = \frac{4}{3}\pi r^3$; $F_{gravity} = \frac{Gm_1m_2}{r^2}$; 1 litre = 1000 cm$^3$; $M_{\text{Earth}} = 5.97 \times 10^{24}$ kg, $\rho(\text{water}) = 1$ g/cm$^3$; $r_{\text{Earth}} = 6.37 \times 10^6$ m

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## Question 1 [8]

The position of a 2 kg object is given as $x(t) = \beta t^2 + 5$, where $x$ is in meters and $t$ is in seconds.

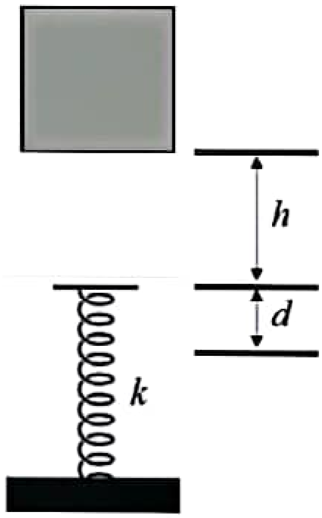
(a) Determine the force $F$ responsible for this motion. (3)

(b) If the force only results in a change in the kinetic energy of the object of 200 J between $t_1 = 0$ s and $t_2 = 5$ s, determine the value of $\beta$. (3)

(c) Without using kinematics (equations of motion), determine the displacement of the object during these 5 seconds referred to in part (b). (2)

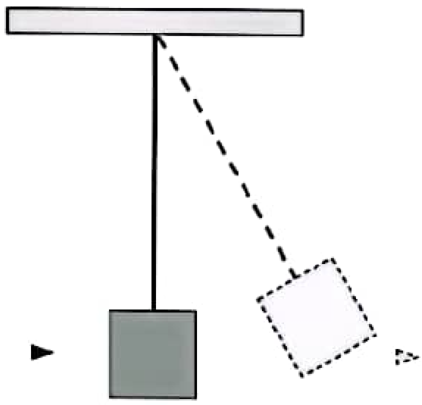
## Question 2 [6]

A spring, with spring constant $k = 1960$ N, is mounted vertically on a table as shown in the figure. A block of mass $m = 2.0$ kg is dropped from a height $h$ and falls on the spring compressing it by a distance $d = 11$ cm. From what height, $h$, above the spring, was the block released from rest?




## Question 3 [12]

A bullet with a speed of 300 m/s passes right through a block with a mass of 400 g. The block is suspended on a long cord. The impulse gives the block a speed of 1.5 m/s.



(a) If the bullet has a mass of 3 g, determine the speed of the bullet after it has passed through the block. (3)

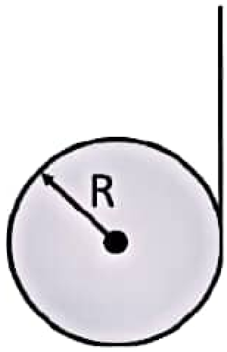
(b) By what vertical distance will the center of mass of the block rise as the block swings upward after the bullet passes through it? (4)

(c) Determine the work done by the bullet as it passes through the block. (2)

(d) How much of the system's mechanical energy (bullet + block) is converted to heat? (3)

4 **Question 4** [9]

A string is wound around a homogeneous, horizontal cylinder of mass $M$ and radius $R$. The cylinder is then held above the ground and released from rest. As the string unwinds and the cylinder spins, the end of the string is continually pulled vertically upwards with a force that is just sufficient to keep the cylinder from descending to the ground. The figure shows the situation viewed from the side.




(a) Identify the forces acting on the system (string and cylinder) and determine an expression for the tension in the vertical portion of the string. (3)

(b) Determine the angular acceleration of the cylinder. (4)

(c) What is the upward acceleration of any point along the vertical portion of the string? (2)

5 **Question 5** [5]

The diaphragm of a loudspeaker oscillates in order to produce sound. The amplitude of such an oscillation has been found to be 5.00 $\mu$m.

(a) At what frequency will the magnitude of the diaphragm's acceleration be equal to $g$. (3)

(b) What will the effect be on the acceleration if the frequency is doubled. Explain your answer. (2)

6 **Question 6** [9]

Two sinusoidal waves in a string are defined by the wave functions

$$y_1 = 2.00 \sin(20.0x - 32.0t)$$

$$y_1 = 2.00 \sin(25.0x - 40.0t)$$

where $x, y_1$ and $y_2$ are in centimeters, $t$ is in seconds and the angles are measured in radians.

(a) What is the phase difference between these two waves at the point $x = 5.00$ cm at $t = 2.00$ s. Give your final answer in degrees between $0^{\circ}$ and $360^{\circ}$. (4)

(b) What is the positive $x$ value closest to the origin for which the two phases differ by $\pm\pi$ at $t = 2.00$ s resulting in a zero displacement? Show all your workings and explain your reasoning. (5)

7 **Question 7** [9]

A lead bullet is fired into a wall and is embedded. A myth claims that a bullet can be completely melted due to friction. A friend of yours doubts that this is possible. Using your physics knowledge you set out to test the validity of the myth. The first assumption that you make is that **ALL** kinetic energy of the bullet is converted into heat due to friction and the second assumption is that it was a pleasant day with an ambient temperature of 25.0$^{\circ}$C. Determine the minimum speed of the lead bullet that will be required to achieve the outcome of the myth. Is this myth of the melting lead bullet plausible if you consider that the average bullet can reach speeds of 700 m/s?
The latent heat of fusion for lead is $2.32 \times 10^4$ J/kg, the specific heat of lead is 128 J/(kg$\cdot^{\circ}$C) and the melting point of lead is 327$^{\circ}$C.

## The end!

## Question 1

a) $x(t) = \beta t^2 + 5$

$v(t) = 2\beta t$

$a(t) = 2\beta$ ✓✓

$F = ma = 2kg(2\beta)$

$= 4\beta$ ✓

b) $v_0 = 0 \text{ m/s}$

$v_5 = 2\beta t = 2\beta(5) = 10\beta$

$\Delta K = 200J = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$ ✓

$= \frac{1}{2}(2kg)(10\beta)^2 - 0$ ✓

$= 100\beta^2$

$\beta = \sqrt{2} = 1.4$ ✓

$\boxed{\text{C)}}$ $W = \Delta K$ ✓

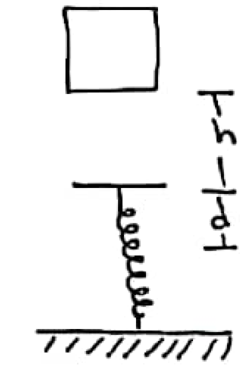
This question specifically stated that you were not to use kinematics. So the correct way to do this problem is to use the Work kinetic energy theorem.

$F\Delta x = \Delta K$

$\Delta x = \frac{\Delta K}{F} = \frac{200J}{4(1.4)N} = 35.4m$ ✓

[8]

## Question 2




— $u_g = 0,\ y = 0$

Define the system:

Block
Spring
Earth

System is isolated.

In energy problems, it is essential that you state the following:
* where the gravitational potential is zero, or where $y = 0$.
* what your system consists of and if it is isolated or not.

Conservation of Energy:

$$\left.\begin{aligned}\Delta E &= \Delta K + \Delta U_g + \Delta U_s + \Delta E_{int} = 0 \\ 0 &= K_f - K_i + mg(y_0) - mg(h+d) + \tfrac{1}{2}kd^2 - \tfrac{1}{2}kx_0^2\end{aligned}\right\}$$

$$= 0 - 0 + 0 - mg(h+d) + \tfrac{1}{2}kd^2 - 0$$

$$h = \frac{-mgd + \frac{1}{2}kd^2}{mg}$$

$$= \frac{-2\,\text{kg}(9.8\,\text{m/s}^2)(0.11\,\text{m}) + \frac{1}{2}(1960\,\text{N/m})(0.11\,\text{m})^2}{2\,\text{kg}(9.8\,\text{m/s}^2)}$$

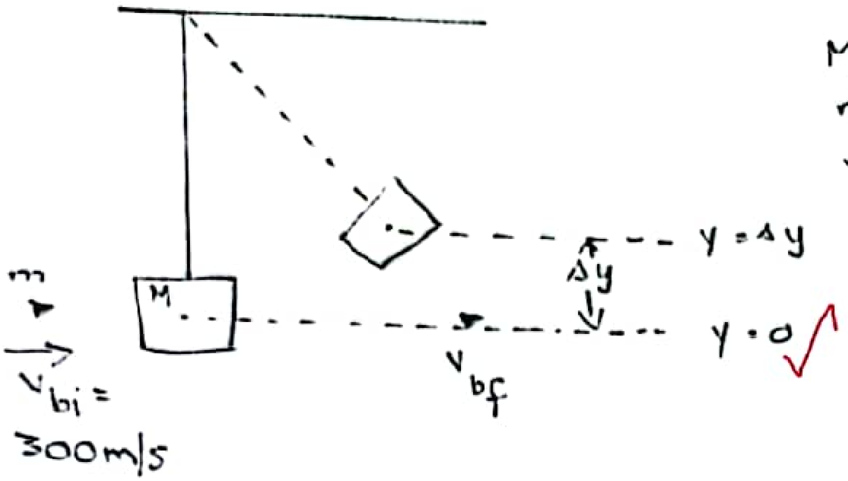
$$\left.\begin{aligned}&= \boxed{0.495\text{ m}} = \boxed{0.50\text{ m}} \\ &= \boxed{49.5\text{ cm}} \approx \boxed{50\text{ cm}}\end{aligned}\right\}$$

[6]

The most common error in this question was that some students did not remember that as the spring compresses, the gravitational potential energy also changes. This is why it is essential to label all the significant points on the diagram, and that you realise that the final position of the block is not at the top of the uncompressed spring, but at the lowest point.

Depending on where you chose y = 0, the substituted values may change, but the final answer will be the same.

## Question 3




$M = 400g = 0.400\,kg$
$m = 3g = 0.003\,kg$
$v_{ib} = 300\ m/s$
$v_{iM} = 0\ m/s$
$v_{fM} = 1.5\,m/s$

a) Conservation of momentum:

$$mv_{bi} + Mv_{Mi} = Mv_{Mf} + mv_{bf}$$

$$v_{bf} = \frac{mv_{bi} + Mv_{Mi} - Mv_{Mf}}{m}$$

$$= \frac{(3\times10^{-3}kg)(300m/s) + (0.400kg)(0 - 1.5m/s)}{3\times10^{-3}\,kg}$$

$$= 100\ m/s$$

b) Conservation of energy
System: Isolated: Block + Earth.

$$\Delta E = \Delta K + \Delta U + \Delta E_{int} = 0$$

$$K_f - K_i + U_{gf} - U_{gi} = 0$$

$$0 - \tfrac{1}{2}mv_i^2 + mgy_f - \cancel{mgy_i}^{=0} = 0$$

$$y_f = \frac{1}{2}\frac{mv_i^2}{mg} = \frac{1}{2}\frac{v_i^2}{g} = 0.115\ m$$

Question 3 cont.

(c) $W = \Delta K$ ✓

$= \frac{1}{2} m_b v_f^2 - \frac{1}{2} m_b v_i^2$

$= \frac{1}{2}(0.003\,kg)((100\,m/s)^2 - (300\,m/s)^2)$

$= -120\,J$ work done on bullet.

Work done by bullet is 120 J. ✓

The work kinetic energy theorem gives you the work done on the bullet, and gives you a negative value for the work done. The work done by the bullet is just the negative of the work done on the bullet - giving you a positive value for the work done.

(d) System: bullet + block.

Initial is just before collision.

Final is just after collision.

$$\left.\begin{array}{l}\Delta E = \Delta K + \Delta U + \Delta E_{int} = 0 \\ \Delta K_b + \Delta K_M + \Delta U_b + \Delta U_M + \Delta E_{int} = 0\end{array}\right\}$$ ✓

$\frac{1}{2} m_b (v_f^2 - v_i^2) + \frac{1}{2} M (v_{fM}^2 - v_{iM}^2) = -\Delta E_{int}$ ✓

$-120\,J + \frac{1}{2}(0.400\,kg)(1.5\,m/s^2 - 0^2) = -\Delta E_{int}$

$\therefore \Delta E_{int} = +120\,J - 0.45\,J = 119.55\,J$ ✓

In the last part of the question, we need to look at the entire system, Bullet, and block. In part c of the question we already calculated the change in the kinetic energy of the bullet, but just after the collision, there is also a change in the kinetic energy of the block which must be taken into account.
For the final answer to be correct - even if you rounded off to 120 J, you need to have shown that the change in the energy of the block was taken into account.

[12]

## Question 4

a) Forces: Weight of cylinder ✓
External Force pulling string. resulting ✓
in Tension


$$\left.\begin{aligned}\Sigma F_y = ma_y = 0 = T - mg\\ T = mg\end{aligned}\right\} ✓$$

For part a - the net force = 0 N, and you need to state Newton's 2nd law and then apply it to the situation.

b) $\tau = \tau_1 + \tau_2 = I\alpha$ ✓

$\alpha = \frac{1}{I}\left[R_1(mg) + R_2 T\right]$ ✓ But $R_1 = 0$

$= \frac{1}{\frac{1}{2}\cancel{M}R_2^{\cancel{2}}}\left[\cancel{R_2}(\cancel{m}g)\right]$ ✓ $R_2 = R = \text{radius}.$

$\alpha = \frac{2g}{R}$ ✓

Here again we need to state Newton's 2nd law for rotation and then apply it to the situation. It is important to realise that although the centre of mass of the cylinder is not accelerating, there is angular acceleration. Using the tension obtained in (a) the acceleration is independent of the mass of the cylinder.

c) $a = \alpha R$ ✓ $= \left(\frac{2g}{R}\right)R$ [sub from (b) ✓]

$a = 2g$ ✓

Even though there is no net acceleration of the cylinder, the string that is unwound from the cyclinder accelerates upwards, otherwise the cylinder would fall to the ground. This acceleration is related to the angular acceleration.

[9]

## Question 5

a) $A = 5.00 \mu m$

For ascillator : $y = A\cos(\omega t + \phi)$

$a = -\omega^2 A \cos(\omega t + \phi)$

$a_{max} = \omega^2 A = g$ ✓

You need to state that this is the maximum acceleration of the diaphragm and relate it to the amplitude of the oscillation.

$\omega = \sqrt{\frac{g}{A}}$

$\omega = 2\pi f = \sqrt{g/A}$ ✓

$f = \frac{1}{2\pi}\sqrt{\frac{g}{A}} = \frac{1}{2\pi}\sqrt{\frac{9.8\, m/s^2}{5\times10^{-6}\, m}}$

$= 223$ Hz ✓

b) $\omega^2 A = a_{max}$

$(2\pi f)^2 A = a_{max}$ ✓

If $f$ is doubled $a_{max}$ will be 4 times ✓ higher

A specific number is needed here - if the frequency is doubled the acceleration will be 4 times larger.
A reason is also needed to substantiate your answer.

[5]

# Question 6

$y_1 = 2.00 \sin(20.0x - 32.0t)$

$y_2 = 2.00 \sin(25.0x - 40.0t)$

a) $\Delta\phi = -(20.0x - 32.0t) + (25.0x - 40.0t)$

$= (5.00x - 8.00t)$

At $x = 5.00$ cm and $t = 2.00$ s

$|\Delta\phi| = 9$ radians $\times \frac{360°}{2\pi \text{ rad}} = 515.6°$

$=$

$\Delta\phi = 155°$

b) $\Delta\phi = (5.00x - 8.00t)$

When $\Delta\phi = (\pm\pi)$ or $(\pm\pi + n2\pi)$ $y_1 + y_2 = 0$.

For $t = 2.00$ s:

$\Delta\phi = (5.00x - 16) = (\pm\pi \pm 2n\pi)$

The critical part of this question is to realise that the phase difference needs to be pi, but the sine function repeats after 2 pi radians. You cannot say that it is just n*pi, because for all even values of n, the waves are inphase.

For $n = 0$; and $+\pi$

$x = 9.57$ cm

For $n = 1$:

$x = 25.4$ cm

increasing

For $+\Delta\phi$ values increase

For $n = 0$, and $-\pi$

$x = 2.57$ cm

For $n = 1$ ($\Delta\phi$ neg.)

$x = 1.32$ cm

$n = 2$

$x = 0.058$ cm

For $-\Delta\phi$ values decrease initially.

## Question 7

Energy stored in the bullet as kinetic Energy

$K = \frac{1}{2}mv^2$

$\Delta K = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$

$= -\frac{1}{2}mv_i^2$ ✓

To melt bullet:

Heat bullet to melting point:

$Q_1 = mc\Delta T$ ✓

~~To~~ Melt

$Q_2 = mL_F$ ✓

If all KE is converted to heat then:

$Q_1 + Q_2 + \Delta K = 0$ ✓

$Q_1 + Q_2 = -\Delta K$

$mc\Delta T + mL_F = \frac{1}{2}mv_i^2$

$2[c\Delta T + L_F] = v_i^2$ ✓

✓

$[2(128\,J/kg\cdot°C)(327°C - 25°C) + 2.32\times10^4\,J/kg]^{1/2} = v_i$

$v_i = 352\,m/s$. ✓

✓

This is plausible, but energy is needed to break the wall in order to embed. ✓

Your final answer needs to be supported by some or other argument for or against the plausibility of the myth